

#if statement:

a=33

b=200

if b>a:

print('b is greater than a')

#else:

a=200

b=33

if b>a:

print('b is greater than a')

else:

print('b is not greater than a')

a=20

b=20

if b>a:

print('b is greater than a')

elif a==b:

print('a and b are equal')

else:

print('a is greater than b')

amount=2000

if amount<1000:

discount=amount*0.05

print("Discount",discount)

elif amount<5000:

```
discount=amount*0.10
print("discount",discount)
else:
    discount=amount*0.15
    print("Discount",discount)
    print("Net payable:",amount-discount)
```

#Nested IF Statements:

```
var=100
if var<200:
    print('Expression value is less than 200')
    if var==50:
        print('which is 150')
    elif var ==100:
        print('which is 100')
    elif var==50:
        print('which is 50')
    elif var<50:
        print('Expression value is less than 50')
    else:
        print('could not find true expression')

print('Good bye!')
```

```
var=185
if var<200:
```

```
print('Expression value is less than 200')

if(var>150 and var<170):

    print('which is greater than 150 less than 170')

elif (var ==100 or var>180):

    print('which is equal to 100 or greater than 180')

elif var==50:

    print('which is 50')

elif var<50:

    print('Expression value is less than 50')

else:

    print('could not find true expression')
```

```
print('Good bye!')
```

```
#while loop:
```

```
count=0
```

```
while(count<9):
```

```
    print('The count is ',count)
```

```
    count=count+1
```

```
print('Good bye!')
```

```
count=0
```

```
while count<7:
```

```
    print(count,'is less than 7')
```

```
    count=count+1
```

else:

print("loop end")

#for loop

range(0,5)

[0,1,2,3,4]

for var in list(range(5)):

print(var)

for i in range(0,30,3):

print(i)

for i in range(0,50,5):

print(i)

#break/continue statement:

for i in range(1,6):

if i==3:

break

print(i)

for i in range(1,6):

if i==3:

continue

print(i)

#pass statement:

for i in range(1,4):

pass

print('Loop completed')

```
x=10

if x>5:

    pass

else:

    print('less than 5')

print('Done')

#Old String formate:

name='john'

print('Hello,%s!'% name)

name='john'

age=23

print('%s %d years old'%(name,age))

#New String format

name='john'

age=23

print('My name is {} and i am {} years old.'.format(name,age))

#named indexes:

txt1="My name is {fname}, I'am {age}".format(fname='john',age=36)

#numbered indexes:

txt2="My name is {0},i'm{1}".format('john',36)

#empty placeholders:

txt3="My name is {},i'm{}".format('john',36)

print(txt1)

print(txt2)
```

```
print(txt3)

#variable declaration

str='java'

str2='C#'

str3='{0}and{1}both are programming languages'.format(str,str2)

print(str3)

#store input numbers:

num1=input('Enter first nmbre:')

num2=input('Enter second number:')

sum=float(num1)+float(num2)

min=float(num1)-float(num2)

mul=float(num1)*float(num2)

div=float(num1)/float(num2)

print('The sum of{0} and {1}is {2}'.format(num1,num2,sum))

print('the subtraction of {0} and {1} is {2}'.format(num1,num2,min))

print('The multiplication of {0} and {1} is {2}'.format(num1,num2,mul))

print('The division of {0} and {1}is {2}'.format(num1,num2,div))

x=input('Enter value of x:')

y=input('Enter value of y:')

temp=x

x=y

y=temp

print('The value of x after swapping:{0}'.format(x))

print('The value of y after swapping:{0}'.format(y))

num=int(input('Enter a number.'))
```

```
if(num%2)==0:
    print('{0} is Even number'.format(num))
else:
    print('{0}is odd number'.format(num))
```

output:

b is greater than a

b is not greater than a

a and b are equal

discount 200.0

Expression value is less than 200

which is 100

Good bye!

Expression value is less than 200

which is equal to 100 or greater than 180

Good bye!

The count is 0

The count is 1

The count is 2

The count is 3

The count is 4

The count is 5

The count is 6

The count is 7

The count is 8

Good bye!

0 is less than 7

1 is less than 7

2 is less than 7

3 is less than 7

4 is less than 7

5 is less than 7

6 is less than 7

loop end

0

1

2

3

4

0

3

6

9

12

15

18

21

24

27

0

5

10

15

20

25

30

35

40

45

1

2

1

2

4

5

Loop completed

Done

Hello, john!

john 23 years old.

My name is john and i am 23 years old.

My name is john, I'am 36

My name is john,i'm36

My name is john,i'm36

javaandC#both are programming languages

Enter first number:2

Enter second number:3

The sum of 2 and 3 is 5.0

the subtraction of 2 and 3 is -1.0

The multiplication of 2 and 3 is 6.0

The division of 2 and 3 is 0.6666666666666666

Enter value of x:2

Enter value of y:3

The value of x after swapping:3

The value of y after swapping:2

Enter a number:2

2 is Even number